Appendix 2 – Detailed Background Research on the Dimensions of Digital Sovereignty

Digital sovereignty operates across multiple interdependent dimensions; layers forming what might be termed a "sovereignty stack": weakness in one dimension constrains capacity in others, while strength in one can compensate for or enable capacity-building in others.

Political Dimension

Authoritarian Digital Sovereignty

China exemplifies comprehensive state control over digital infrastructure and information flows. The "Great Firewall" employs technical filtering, DNS manipulation, and protocol-level blocking to enforce territorial boundaries in cyberspace (King, Pan, and Roberts 2013; Fleming 2025). The Cybersecurity Law (2017) and Data Security Law (2021) mandate data localization and grant authorities broad surveillance powers. This architecture creates what the government calls "cyber sovereignty": each state's right to control internet content and data within its borders, justified as protecting political stability and cultural values against foreign influence.

Russia's "sovereign internet" law (2019) similarly aims to reroute traffic domestically and isolate its network segment if threatened externally. Both examples demonstrate how states deploy technical architecture to enforce political control: filtering technologies (design) constrain information access (affordance), enabling content governance (sovereignty). Yet these systems also reveal sovereignty's costs, such as economic inefficiencies from isolation, reduced innovation from censorship, and intensified social control that many citizens resist through VPNs and circumvention tools.

Democratic Regulatory Sovereignty

The European Union pursues digital sovereignty through regulatory frameworks balancing openness with rights protection. The General Data Protection Regulation (2016) established enforceable privacy rights and data portability requirements. The Digital Markets Act (2022) and Digital Services Act (2023) regulate platform market power and content moderation, requiring interoperability and transparency. The AI Act (2024) creates risk-based governance for artificial intelligence systems (Fleming 2025). These instruments assert regulatory sovereignty: the EU's power to set rules for any company operating in European markets, regardless of headquarters location.

This "Brussels Effect" demonstrates how large regulatory jurisdictions can establish de facto global standards (Bradford 2023). When platforms redesign systems to comply with EU law, those changes often apply worldwide because maintaining separate architectures is costlier than universal compliance. However, regulatory sovereignty confronts limits when enforcement depends on foreign corporations, when rules apply only to conduct within territorial boundaries that data flows routinely cross, or when technical capacity to audit and verify compliance is absent.

Global South Techno-Nationalism

India illustrates a third model combining elements of both authoritarian control and democratic aspiration. The government banned dozens of foreign applications on national security grounds, promoted data localization through draft regulations, and positioned data as a national asset requiring sovereign control. India's approach reflects techno-nationalist framing: digital sovereignty as reducing dependence on foreign technology providers while building domestic capacity. Similar dynamics appear across the Global South, where digital sovereignty rhetoric addresses legitimate concerns about neo-colonial data extraction and technological dependency while potentially enabling surveillance and protectionism.

Other countries in Latin America and Africa emphasize access and closing the digital divide as a sovereignty issue, sometimes partnering with foreign firms due to limited domestic capacity (raising concerns of new dependencies) (Ricaurte 2019; Gagliardone 2020).

Yet these sovereignty assertions often lack the technical and economic capacity to implement alternatives. Banning foreign platforms without viable domestic replacements leaves users with fewer options. Data localization requirements without local cloud infrastructure force dependence on foreign providers operating data centers domestically, creating sovereignty theater rather than genuine autonomy. Effective sovereignty requires regulatory assertion combined with institutional capacity, technical expertise, and economic resources to build alternatives.

Sovereign AI and Corporate Partnerships

Recent developments around "sovereign AI" reveal evolving political-corporate sovereignty entanglements. When OpenAI released ChatGPT in November 2022, governments worldwide immediately recognized that control over foundational AI capabilities could determine economic competitiveness, national security, and cultural influence for decades. Within months, nations rushed to assert technological sovereignty through what became known as the "sovereign AI race" (Chavez 2024).

This competition took multiple forms across 2023-2024. Singapore invested $52 million in SEA-LION, training models on 11 Southeast Asian languages to counter what developers explicitly called "West Coast American bias" in dominant English-centric systems. Taiwan launched TAIDE with $7.4 million specifically to counter Chinese AI chatbots adhering to "core socialist values" — a direct assertion of cultural and political sovereignty through AI architecture. The Netherlands invested €13.5 million in GPT-NL as public interest AI embodying European values, with 95% profit-sharing provisions ensuring community benefit. Japan's Fugaku project aimed for large language models not relying on foreign technology, pursuing complete technical self-sufficiency (Chavez 2024).

Beyond training models, nations recognized that AI infrastructure sovereignty requires control over computational resources. France invested $44 million in Jean Zay supercomputer equipped with 1,456 Nvidia H100 GPUs, Japan allocated $730 million for AI computing infrastructure, Australia committed $35 million to sovereign AI compute capacity, and the European Union allocated roughly $2 billion for "AI Factories" through its EuroHPC initiative (Chavez 2024). These investments aimed to reduce dependence on American and Chinese cloud providers whose terms of service and data practices raised sovereignty concerns (Bradford 2023).

OpenAI's partnerships with non-democratic governments, including the United Arab Emirates, signal normalization of authoritarian technology collaboration. OpenAI executive Jason Kwon's justification, "engagement is better than containment", echoes President Clinton's 2000 argument for integrating China into the global economy despite human rights concerns. This contrasts sharply with 2019, when Google employees successfully resisted the company's Project Dragonfly collaboration with Chinese authorities. The shift suggests corporate sovereignty claims increasingly override earlier norms about limiting authoritarian technology access.

These partnerships are marketed as enabling "AI sovereignty": national capacity to develop and deploy artificial intelligence without foreign dependence. Yet when sovereignty requires licensing models from foreign corporations, operating on their infrastructure, and accepting their terms, the arrangement resembles dependence more than autonomy. States gain limited capabilities while corporations expand markets and influence. True AI sovereignty would require independent capacity to train models, access to computational resources (addressing Nvidia's 92% GPU market share chokepoint), and domestic expertise; conditions few nations possess (Farooque 2025).

Other Political Actors

When private corporations control the arenas in which citizens debate, organize, and access essential services, they wield a kind of power that operates beneath formal political institutions yet profoundly affects democratic capacities (Cohen 2019; Zuboff 2020). Thus, the political dimension of digital sovereignty is also a battle to subject these private authorities to public oversight or to replace them with more accountable alternatives.

Civil society and social movements form a third force in this contest, often aligning with neither states nor corporations. Digital rights activists push back against state surveillance and corporate data abuse, advocating for user-centric governance of the internet. They promote multi-stakeholder venues for policymaking (where civil society has a seat), and experiment with decentralized networks that operate outside both state and corporate control (like peer-to-peer mesh networks during protests to evade shutdowns) (DeNardis and Raymond 2013; Milan and Hintz 2013; Mueller 2017).

This dynamic can be seen as a triangular struggle. For example, in the realm of content moderation: states invoke sovereignty to demand platforms remove certain content (hate speech, "fake news") under local law; platforms weigh these requests against their own rules and global reputations; and civil society monitors both, sometimes litigating against state censorship and sometimes calling out platforms for amplifying harmful content (Gillespie 2018; Gorwa 2019).

Companies worry that rampant state interventions (different rules in every country) could balkanize the internet and hurt their business. Civil society worries that both unchecked corporate algorithms and unchecked state surveillance threaten democracy. This shared unease has prompted proposals for new governance models. For instance, the idea of a

“Digital Geneva Convention” (floated by Microsoft) seeks a sovereign-level pact to protect civilians from cyberattacks, analogous to war conventions (Smith 2017).

Underlying these struggles is a sense that existing political frameworks are lagging behind the pace of digital change. Internationally, there is no consensus treaty fully addressing digital rights or governance. Attempts at global coordination, such as the United Nations’ Internet Governance Forum (IGF) or the more recent proposal for a “Global Digital Compact”, highlight a desire for common principles, but geopolitical rivalries complicate progress (United Nations 2023).

Economic Dimension

Platform Capitalism and Surveillance Capitalism

Platform business models concentrate economic power by controlling infrastructure through which others must operate. Platforms function as digital landlords, extracting rent by positioning themselves between buyers and sellers, workers and customers, advertisers and audiences (Srnicek 2016). Amazon controls retail infrastructure, taking commissions on third-party sales while using transaction data to compete with successful merchants through its own brands. Google and Meta dominate digital advertising, capturing revenue that once supported journalism and creative industries. Apple and Google control mobile app distribution, imposing fees and rules on developers accessing their ecosystems.

This architectural power enables what Zuboff (2019) terms "surveillance capitalism": converting human experience into behavioral data that fuels prediction markets. Every search query, social media interaction, location movement, and purchase becomes raw material for algorithms predicting and influencing future behavior. Users generate data through ordinary activities; platforms appropriate it without compensation; advertisers and other clients purchase predictions. The economic sovereignty implication is stark: users function as raw material, with no say over data use or value distribution.

The result is a massive transfer of value: for example, African users generate data that fuels AI models owned by Western firms, or gig workers in Southeast Asia contribute to apps whose profits flow to Silicon Valley, leaving their communities only precarious incomes (Kwet 2019; Ricaurte 2019). This concern underpinned India's and Indonesia's early talk of data localization: a desire to keep data domestically to spur local data-driven innovation and prevent all value from accruing abroad.

There is a burgeoning movement advocating economic alternatives grounded in cooperative and commons-based principles. This is where concepts like the digital commons, platform cooperatives, and community wealth building come in (Bauwens, Kostakis, and Pazaitis 2019; Scholz 2023). The premise of these initiatives is that the fruits of the digital economy (data, creative content, software, even algorithms) can be produced and managed as shared resources rather than proprietary assets. Platform cooperatives offer alternative ownership models where workers and users collectively govern platforms and share value (Scholz 2023). Examples include Stocksy (photographer-owned stock image platform), Resonate (musician-owned streaming service), and CoopCycle (delivery worker-owned logistics platform). Similarly, data cooperatives and trusts are being proposed to let individuals pool their data and negotiate its use collectively, potentially earning dividends or ensuring it is used for social good (e.g. medical research) (Hafen, Kossmann, and Brand 2014; Delacroix and Lawrence 2019). These models aim to re-embed technology within relationships of solidarity and mutual benefit. However, they struggle against platform monopolies' network effects, capital access advantages, and willingness to operate at losses subsidized by investors expecting future monopoly rents.

Twin Extractivism: Intangibles and Nature

Digital capitalism operates through what Rikap (2021) theorizes as "twin extractivism": the simultaneous appropriation of intangibles (data, knowledge, attention) and nature (energy, water, minerals). This dual extraction reveals the inseparability of digital and ecological sovereignty. Data centers require massive material infrastructure: constant electricity consumption, water for cooling systems, rare earth minerals for hardware manufacturing. Yet the value these infrastructures generate flows primarily to platform owners and shareholders, not to communities providing resources or generating data.

Consider data centers' economic footprint. A typical hyperscale facility employs approximately 20 people while operating 24/7 with stadium-sized buildings consuming electricity and water at scales rivaling small cities (Marx 2025). Host regions are promised economic development but receive minimal employment and often surrender environmental protections to attract investment. Latin America, for instance, produces less than 20% of the material inputs required for data centers built on its territory, importing most components while exporting resources and providing cheap energy (Rikap et al. 2024). This pattern mirrors historical extractive relationships where peripheral regions supply raw materials while core regions capture value-added production and profits.

Chile's experience illustrates twin extractivism's operation. When the government developed a sustainability assessment tool for data center siting, industry pressure led authorities to raise toxicity thresholds, exempting data centers from environmental reporting requirements that applied to other industries (Marx 2025).

Data Colonialism and Global Inequalities

Data colonialism describes how historical patterns of extraction and appropriation continue through digital means (Couldry and Mejias 2019). Global South populations generate data consumed by Northern artificial intelligence systems, yet lack infrastructure or capacity to develop their own technological alternatives. Languages, cultural practices, and local knowledge are datafied and monetized by foreign corporations without consent, compensation, or community benefit. This represents appropriation of collective knowledge as private property, enclosing commons through technical and legal mechanisms.

The AI training data economy exemplifies these dynamics. Foundation models powering ChatGPT, Claude, and similar systems train on text scraped from across the internet, including content from Global South creators who receive no compensation and whose languages remain underrepresented in resulting systems. When these models are deployed globally, they encode Western cultural assumptions and English-language dominance, marginalizing other knowledge systems while extracting their raw materials.

Moreover, the "sovereign AI race" discussed in the political dimension has economic dimensions. Nations investing in domestic AI capabilities confront fundamental dependency on semiconductor supply chains dominated by few companies. Nvidia's 92% GPU market share creates an inescapable chokepoint: even "sovereign" AI systems train on foreign hardware subject to export controls and corporate terms (Farooque 2025). True economic sovereignty would require domestic semiconductor manufacturing, energy infrastructure, and technical expertise. These are conditions that decades of neoliberal globalization deliberately dismantled in most nations.

Value Extraction and Labor

Digital platforms restructure labor relations in ways that concentrate value while fragmenting worker power. Algorithmic management systems survey, measure, and discipline workers with precision impossible under previous arrangements. Platform workers (ride-hail drivers, delivery couriers, content moderators, data annotators) often misclassified as independent contractors lack benefits, job security, or collective bargaining rights, yet face unilateral rule changes, opaque performance metrics, and arbitrary deactivation (Graham et al. 2020).

This restructuring enables global labor arbitrage. Content moderation for Facebook happens primarily in Philippines and Kenya, where workers view traumatic material for low wages under intense productivity quotas (Soriano 2021). Data annotation for machine learning systems employs workers worldwide through platforms like Amazon Mechanical Turk, paying piece rates below minimum wage. The cloud thus enables corporations to disaggregate production, accessing labor globally while avoiding local employment regulations. It is a key sovereignty challenge when work crosses borders through platforms operating under terms workers cannot negotiate.

Worker organizing confronts architectural obstacles. Platforms prevent drivers from communicating with each other, hide algorithms determining pay and assignments, and terminate workers who organize collectively. Yet resistance emerges through various forms: worker cooperatives building alternative platforms, unions advocating for employment reclassification, and solidarity networks sharing information despite platform constraints. Economic sovereignty for workers requires both regulatory intervention compelling platforms to recognize labor rights and technical capacity to build worker-owned alternatives (Rosenblat 2018; Wood et al. 2019).

State Capture

The United Kingdom's five-year Microsoft agreement describes lucidly how procurement decisions can constitute sovereignty surrender (Marx 2025). Signed immediately after appointing Microsoft's UK CEO to a key industrial strategy role, the deal locks government departments into cloud dependency without developing domestic alternatives. When competition authority chairman Sarah Cardell resigned and was replaced by former Amazon UK manager Doug Gurr, the pattern became clear: revolving doors between corporate executives and government positions facilitate corporate sovereignty over state institutions.

This arrangement is recurring across geographies, and delivers minimal economic benefit while creating deep dependency. Public sector data flows to foreign corporations whose terms of service and data practices the state cannot control. Public funds subsidize private monopolies rather than building public infrastructure that could recirculate value locally (Rikap et al. 2024). The sovereignty implication extends beyond technology to democratic governance itself: when former industry executives regulate their former employers and award procurement contracts, corporate capture supplants public accountability.

Strategic Responses

An essential aspect of economic sovereignty is the idea of self-sufficiency or resiliency. The COVID-19 pandemic underscored the dependence many economies have on a few cloud computing providers and supply chains for hardware. This has spurred calls for investing in local tech capacity to avoid being economically at the mercy of foreign powers or companies for key digital infrastructure. For developing countries, complete self-sufficiency may be unrealistic, but regional alliances or South-South technology cooperation are being explored. For example, several Latin American countries have discussed developing regional data centers or an alternative fiber optic backbone to reduce reliance on U.S. infrastructure. Likewise, the African Union's Digital Transformation Strategy emphasizes building continental platforms and encouraging African startups (Brevini, Hintz, and McCurdy 2013; African Union 2018; Soriano 2021).

Reclaiming economic sovereignty involves multiple policy approaches. Governments deploy antitrust actions limiting platform monopolies, taxation measures (digital services taxes capturing foreign tech revenue), and requirements for local reinvestment or partnerships (Cockfield and MacArthur 2015; Khan 2017; Smith, Monea, and Santiago 2022). Another tool is data localization and national data governance policies. For example, India's draft

Personal Data Protection law at one point proposed strict data localization, partly to force foreign companies to store data in India, thereby subjecting it to Indian jurisdiction and potentially enabling domestic alternatives to develop (Rajmohan 2025). Countries like Nigeria have demanded that platforms register local entities and pay taxes, and some nations are exploring treating certain types of data (like telecom data or health data) as a national resource. Critics argue that if taken too far, such policies can backfire (e.g. raising business costs or prompting retaliation), but proponents see them as necessary to prevent a one-way flow of data wealth.

Governments also nurture local digital ecosystems through startup incubators, AI research hubs, and manufacturing investment. China built indigenous platform analogues (WeChat, Alibaba) (Jia and Winseck 2018); smaller nations pursue niche strengths like Estonia's e-governance (Anthes 2015) or Kenya's mobile money innovation (Morawczynski and Pickens 2009; Mohapatra 2011). The goal: homegrown champions abiding by local rules and retaining profits domestically.

Technical Dimension

Infrastructure Control and Dependencies

Digital infrastructure ownership concentrates in few corporate and state hands, creating chokepoints that constrain sovereignty for all other actors. Three technology companies (Amazon Web Services, Microsoft Azure, and Google Cloud) control approximately 65% of global cloud computing capacity (Synergy Research Group 2025). Governments, hospitals, universities, and businesses worldwide run critical operations on infrastructure owned by these firms, subject to their terms of service, pricing structures, and implicit geopolitical considerations.

Undersea cables carrying 95% of international data traffic belong primarily to technology corporations and telecommunications companies. Google alone owns or leases capacity on cables spanning the globe, giving the company privileged access to data flows and leverage over internet connectivity that rivals state power (Starosielski 2015). When cables route through certain jurisdictions, those states gain surveillance and control opportunities. Infrastructure geography thus determines jurisdictional reach: where data physically transits implies which governments can access, intercept, or regulate it.

Wireless spectrum allocation represents another infrastructure domain where sovereignty operates. States license spectrum to telecommunications providers, but corporate consolidation means a handful of companies control mobile connectivity globally. The transition to 5G intensified geopolitical competition, with the United States pressuring allies to exclude Chinese equipment manufacturer Huawei from networks, citing national security concerns. The dispute reveals infrastructure's dual character: simultaneously technical (which equipment provides optimal performance) and political (whose infrastructure grants surveillance or sabotage capabilities) (Kim, Lee, and Kwak 2020).

Hardware Chokepoints

Semiconductor manufacturing creates fundamental technical dependencies that constrain sovereignty claims across all domains. Taiwan Semiconductor Manufacturing Company (TSMC) fabricates approximately 60% of the world's semiconductors and 90% of the most advanced chips (Center for Security and Emerging Technology et al. 2021). Any disruption to TSMC production would cripple global digital systems within months. This concentration explains why the United States allocated $52 billion through the CHIPS Act (2022) for domestic semiconductor manufacturing, pursuing technical sovereignty through industrial policy (The White House 2021, 2022).

Artificial intelligence systems face an even more severe chokepoint. Nvidia controls 92% of the GPU market for AI training, creating inescapable dependency for any actor pursuing machine learning capabilities (Schiffer and Matzakis 2025; Farooque 2025). Even "sovereign AI" initiatives training models domestically operate on Nvidia hardware subject to U.S. export controls and corporate licensing terms. When Meta releases open-source models like Llama, they train on Nvidia infrastructure, meaning supposed openness still depends on proprietary hardware controlled by a single corporation (Touvron et al. 2023).

This hardware sovereignty challenge extends beyond manufacturing to rare earth minerals required for chip production. China controls approximately 90% of rare earth processing capacity, despite holding only 37% of global reserves (US-China Economic and Security Review Commission 2025). Western nations' dependence on Chinese rare earth supply became apparent when China threatened export restrictions during trade disputes. True technical sovereignty would require secure access to material inputs, dependent on factors like geology, industrial capacity, and geopolitical relationships.

Protocols and Standards

Technical protocols function as constitutional infrastructure, establishing rules for how systems interact and who can participate. Open protocols like TCP/IP, HTTP, and SMTP enabled the internet's decentralized growth, allowing anyone to connect without seeking permission from central authorities. Closed protocols, conversely, concentrate control: only those authorized by protocol owners can develop compatible systems or access networks.

Internet standards development happens primarily through multi-stakeholder institutions like the Internet Engineering Task Force (IETF) and World Wide Web Consortium (W3C), where technical experts, corporations, academics, and civil society representatives negotiate protocols (DeNardis 2015). However, corporate influence within these bodies has grown. Large technology companies dedicate substantial resources to standards development, shaping protocols to align with their business interests. When Google proposes new web standards that Chrome browser can implement immediately, its market dominance (65% browser share) effectively makes those standards mandatory for websites wanting to reach users (Hu et al. 2024).

Proprietary APIs (Application Programming Interfaces) represent another standards domain where corporate sovereignty operates. When platforms like Twitter or Facebook restrict API access, they determine which third-party developers can build compatible applications and on what terms. API design choices structure what actions are possible, easy, or prohibited, functioning as architectural legislation that establishes rules without democratic deliberation (Gillespie 2018).

Many current innovations, like distributed ledger technologies (e.g. blockchains) or peer-to-peer protocols, claim to enhance sovereignty by eliminating centralized intermediaries (Greenfield 2017; DuPont 2019). A blockchain like Bitcoin or Ethereum allows users to transact or run applications on a network not controlled by any single authority. This has spurred experiments in decentralized autonomous organizations (DAOs) which manage resources or decision-making through code and collective consensus, aiming to bypass traditional corporate or state hierarchies (Santana and Albareda 2022).

Open-source software offers an alternative model where code remains publicly inspectable and modifiable. Projects like Linux, Firefox, and Signal demonstrate that communities can develop sophisticated systems outside corporate control. Yet open-source development faces sustainability challenges: most projects depend on volunteer labor or corporate sponsorship, creating precarity and potential capture (Eghbal 2020). When corporations employ core developers, they gain de facto influence over project direction even in nominally open systems.

Self-Hosting and Decentralization

Technical sovereignty at individual and community scales increasingly involves self-hosting: running one's own servers and services rather than depending on corporate cloud infrastructure. Tools like Nextcloud (file storage), Mastodon (social networking), and Matrix (messaging) enable individuals and communities to operate digital infrastructure they control (Gehl 2025). Hardware projects like Start Nine, Umbrel, and FreedomBox package self-hosting software with user-friendly interfaces, reducing technical barriers to entry.

However, self-hosting confronts significant obstacles. Most internet service providers prohibit running servers on residential connections, forcing users onto business plans or dedicated hosting. Dynamic IP addresses and port blocking make home servers difficult to access remotely. Email self-hosting proves particularly challenging because major providers (Gmail, Outlook) often reject mail from independent servers as potential spam, effectively enforcing centralization through technical discrimination.

Even when individuals overcome these barriers, self-hosting devices and data remain vulnerable to surveillance and seizure. Without robust encryption and secure hardware (features often absent from consumer devices), self-hosted systems offer limited protection. True technical sovereignty thus requires control extending from application layer through operating system to hardware; a complete stack few actors can achieve independently (The Investor's Podcast Network 2025).

State Technical Capacity

Governments increasingly recognize that regulatory sovereignty requires technical capacity to audit systems, verify compliance, and develop alternatives. The European Union's AI Act mandates risk assessments and algorithmic transparency, but enforcement depends on technical expertise to examine model architectures, training data, and inference processes. When regulators lack this capacity, they must rely on corporate self-reporting or third-party auditors (Raji et al. 2020).

Some governments invest in building public technical infrastructure as sovereignty strategy. Barcelona's municipal data commons, built on open-source platforms with local technical capacity, demonstrates how cities can assert control over digital systems serving residents (Calzada 2018). Estonia's e-governance infrastructure, developed domestically with open standards and data portability requirements, enables citizens to control personal information while government services interoperate seamlessly (Anthes 2015). These examples show technical sovereignty as buildable through sustained investment in public infrastructure and expertise.

Legal Dimension

Jurisdictional Complexity

Traditional legal sovereignty assumes territorial boundaries define jurisdictional reach: states make laws within borders, courts adjudicate disputes among parties present in territory, and enforcement mechanisms operate within geographic limits. Digital systems disrupt these assumptions. When a French citizen uses a U.S. platform to communicate with a Japanese friend, storing data on servers in Ireland while the company's headquarters sits in California, which jurisdiction's laws apply?

This question has generated competing jurisdictional theories. The effects doctrine holds that states may regulate conduct occurring elsewhere if it produces effects within their territory (Bradford 2023). The European Union employs this principle: GDPR applies to any company processing EU residents' data, regardless of where the company operates or where processing occurs (European Commission 2016; Bradford 2023). Conversely, the targeting doctrine focuses on whether conduct deliberately targets users in a jurisdiction, limiting regulatory reach to intentional engagement with local markets (Goldsmith and Wu 2006). China's cyber sovereignty approach emphasizes territorial data sovereignty: data about Chinese citizens or generated within China must be stored domestically and subject to Chinese law, asserting jurisdiction through data localization requirements (Shen 2016; Creemers 2020).

None of these doctrines resolves jurisdictional conflicts when different states claim authority over the same digital activity. When a platform removes content to comply with one state's censorship demands, it may violate another state's free expression protections. When data localization requirements force companies to store information separately in multiple jurisdictions, compliance costs may exclude smaller actors from global markets, entrenching dominant platforms. States assert control by making compliance so burdensome that only large corporations can navigate the complexity, inadvertently reinforcing the power they seek to constrain.

There is an emerging concept of 'sovereignty in cyberspace' as a principle of international law: whether and how the notion of territorial sovereignty applies to state actions in the digital realm (such as hacking into another country's servers). This debate will shape the norms of state behavior; effectively, whether digital sovereignty will be respected between nations (Schmitt 2017; Efrony and Shany 2018).

Data Protection and Privacy Rights

The European Union's General Data Protection Regulation (2016) represents the most comprehensive assertion of legal sovereignty over personal data. GDPR establishes enforceable rights: individuals can access data held about them, correct inaccuracies, delete information under certain conditions, and port data to competing services (European Parliament 2016). These rights create affordances for personal data control, though their effectiveness depends on technical implementation and enforcement capacity.

GDPR's extraterritorial reach applies to any entity processing EU residents' data. The so-called “Brussels Effect” occurs when companies redesign systems globally to comply with EU law because maintaining separate architectures for different jurisdictions proves costlier than universal compliance (Bradford 2023). Yet this sovereignty remains incomplete. Enforcement depends on companies implementing technical measures (data portability tools, deletion mechanisms) that they control, on regulators possessing resources to investigate violations across complex technical systems, and on political will to impose meaningful penalties rather than accepting symbolic fines.

Other jurisdictions have followed Europe's lead while adapting frameworks to local priorities, indicating a broader legal movement to recognize data protection as a right. California's Consumer Privacy Act (2018) and Brazil's Lei Geral de Proteção de Dados (2020) establish data protection rights, though with different scopes, enforcement mechanisms, and philosophical foundations (Greenleaf 2021). China's Personal Information Protection Law (2021) grants individual rights while subordinating them to state security interests and requiring companies to facilitate government access (Sun 2024).

Cross-Border Data Flows

International data transfers represent a critical sovereignty challenge. The Schrems II decision (Court of Justice of the European Union 2020) invalidated the EU-U.S. Privacy Shield framework, ruling that U.S. surveillance laws provide inadequate protection for European data transferred to American companies. The court found that U.S. national security authorities' broad access to data contradicts EU fundamental rights, making American companies unable to guarantee GDPR compliance for data they control.

The ruling created immediate practical difficulties. Thousands of companies transferring data to the United States for processing, storage, or analytics faced legal uncertainty. Standard Contractual Clauses (SCCs) — legal agreements companies sign to enable transfers — remained valid but required case-by-case assessments of whether destination jurisdictions provide adequate protection (European Commission 2021). Companies must now evaluate whether foreign governments can access transferred data and implement additional safeguards like encryption or pseudonymization to protect information from surveillance (Fox 2019). This places legal obligations on companies that they sometimes cannot satisfy.

Data localization requirements offer an alternative approach but generate their own sovereignty tensions. When countries mandate that data about citizens or data generated domestically must be stored within territorial borders, they assert control over information flows. India, Russia, and Vietnam have implemented various localization requirements, justified as protecting privacy, ensuring law enforcement access, and building domestic technology capacity (Chander and Lê 2015; Rajmohan 2025). Critics argue localization fragments the internet, raises costs, and fails to protect privacy when authoritarian governments gain easier access to locally stored data. The debate reveals competing sovereignty visions: networked data flows enabling global services versus territorial control protecting local interests.

Individual versus Collective Rights

Most data protection frameworks prioritize individual rights: personal control over one's own information. GDPR's consent requirements, access rights, and deletion provisions treat data protection as individual prerogative. Yet this individualistic approach confronts limitations when data's value and harms emerge collectively. Algorithmic systems discriminate against demographic groups, not isolated individuals. Population-level health data generates insights no individual dataset could produce.

Current legal frameworks struggle to recognize collective data rights. Property law treats data as either corporate assets or personal information belonging to individuals, with limited categories for communal ownership (Cohen 2019). Contract law structures data governance through individual consent rather than collective deliberation (Viljoen 2021). Competition law addresses market concentration but not whether platform governance respects community autonomy.

These limitations have prompted calls for collective data rights and governance mechanisms. Data trusts allow communities to pool data and negotiate its use collectively, potentially earning benefits or ensuring socially beneficial purposes (Delacroix and Lawrence 2019). Data cooperatives enable members to govern data according to shared values rather than individual transactions (Hafen, Kossmann, and Brand 2014). Indigenous data sovereignty asserts community authority over data about peoples, lands, and knowledge systems, rejecting individualistic frameworks that conflict with collective decision-making traditions (Walter et al. 2021).

New Zealand, in partnership with Māori communities, has begun incorporating principles of Māori data governance into its statistics law and data strategies, acknowledging that customary laws and collective stewardship should influence how data is handled (Hudson et al. 2020). The notion of 'community data' is surfacing in legal debates, with countries like India considering if certain datasets (like anonymized public health data) should be treated as a societal commons rather than owned by whoever collects it (Purtova 2017; Biega et al. 2020). The ILO has even discussed data rights for workers as a group (e.g., drivers collectively owning their ratings data on a platform) (De Stefano 2018). These legal innovations are nascent but indicate an attempt to reconcile Western individual-rights frameworks with communal concepts of sovereignty.

Human rights law overlay this, serving as a check on sovereignty. Sovereignty traditionally allows wide berth for domestic law, but human rights treaties and customary law constrain states from violating fundamental rights even in asserting sovereignty. For example, a government may claim “information sovereignty” to justify an internet shutdown during protests, but human rights bodies (UN Human Rights Council, regional courts) have deemed many such shutdowns disproportionate and thus a breach of international law. Similarly, mass surveillance by states is increasingly challenged as violating the right to privacy. Thus, legal sovereignty is not absolute online any more than offline; it exists within a web of human rights commitments.

Formal international law-making's slowness has spurred soft law alternatives: tech ethics guidelines, industry codes, and declarations like OECD's AI Principles establish non-binding normative baselines that domestic laws may later incorporate (OECD 2019). The United Nations has affirmed that offline rights (privacy, freedom of expression) must be protected online, with UN Special Rapporteurs increasingly scrutinizing government and corporate practices for human rights compliance in digital contexts (United Nations Human Rights Council 2016; Khan 2025). While enforcement is often lacking, this transnational legal discourse pressures sovereigns to justify their digital policies in universalist terms.

Enforcement Challenges

Legal sovereignty confronts enforcement gaps when jurisdiction and capacity diverge. States may claim authority to regulate foreign platforms but lack mechanisms to compel compliance beyond blocking access (a blunt tool often politically costly). Courts may issue judgments but cannot easily enforce them against companies with no physical presence in the jurisdiction. Regulators may impose fines but collection proves difficult when companies hold assets elsewhere and can restructure to limit liability.

These challenges explain why legal sovereignty increasingly operates through design mandates rather than conduct prohibitions. Instead of trying to control what companies do with data after collection, GDPR requires privacy-by-design: building protection into systems from the outset (Article 25). The Digital Markets Act mandates interoperability: requiring platforms to enable competitors' access rather than prohibiting anticompetitive conduct after the fact (Smuha 2025). The AI Act imposes risk assessments and documentation requirements: demanding transparency through technical architecture rather than relying on external oversight alone (European Parliament 2024).

Unfortunately, design mandates depend on technical capacity to specify requirements, verify compliance, and audit implementations; capacities states often lack. Legal sovereignty thus circles back to technical sovereignty: the ability to understand, evaluate, and, if necessary, build alternative systems.

Social Dimension

Digital Divides: Access and Meaningful Use

Digital sovereignty requires more than connectivity. While approximately 67% of the global population now has internet access, this aggregate figure obscures persistent inequalities (International Telecommunication Union 2024). Sub-Saharan Africa's 37% connectivity rate contrasts sharply with Europe's 89%, and within countries, urban-rural divides, income disparities, gender gaps, and age differences structure who accesses digital systems and on what terms. The "digital divide" thus describes intersecting exclusions across axes of marginalization.

Moreover, access alone does not constitute sovereignty. Connecting marginalized communities to corporate platforms controlled elsewhere, on terms they cannot influence, extends digital colonialism. When connectivity requires accepting surveillance, when devices lock users into proprietary ecosystems, when platforms impose languages and cultural norms that marginalize local practices, access becomes a vehicle for subordination.

The affordances that digital systems provide vary dramatically by user position. A software developer in Silicon Valley encounters technology as malleable material subject to modification and control. A rural farmer in Karnataka accesses the same internet through a device they cannot repair, running software they cannot inspect, generating data appropriated without consent or compensation. An elderly person confronts interfaces designed for young users, while people with disabilities navigate systems that assume able-bodied interaction.

Design choices that seem neutral to privileged users function as exclusions for others, demonstrating how sovereignty operates through differential affordance distribution.

Cultural Sovereignty and Knowledge Systems

Digital systems encode particular ways of knowing and being that marginalize alternatives. Artificial intelligence models train predominantly on English-language text, incorporating Western cultural assumptions, philosophical frameworks, and social norms as universal (Bender et al. 2021). UNESCO reports that 43% of the world's approximately 6,700 languages face endangerment, yet AI systems accelerate linguistic concentration by making only a handful of languages computationally viable (UNESCO 2021). When foundation models powering translation services, search engines, and communication platforms operate primarily in English, Mandarin, and Spanish, they structure which languages can participate in digital futures.

This linguistic sovereignty challenge extends to epistemology. Many Indigenous languages encode relationships to land, kinship structures, and temporal orientations that dominant languages cannot express adequately. When digital systems require translation for computational processing, they force knowledge into alien categories, erasing meanings that cannot survive translation. Language is a knowledge system, and linguistic sovereignty involves preserving ways of knowing that algorithmic systems built on dominant languages threaten to erase.

Beyond algorithmic systems, digital platforms reshape cultural production and circulation. Social media platforms privilege particular content forms (short videos, viral memes, algorithmically optimized posts) over others (long-form writing, complex arguments, slow cultural work), restructuring what cultural expression becomes visible and valued (Dijck, Poell, and Waal 2018). Musicians, writers, and artists worldwide increasingly depend on platform algorithms for audience reach, yet these algorithms optimize for engagement metrics that often conflict with cultural or artistic values. Indigenous storytelling traditions emphasizing context, relationship, and ceremonial protocol struggle to translate into platforms designed for rapid sharing and individual authorship claims.

Digital archiving and cultural heritage preservation present sovereignty challenges when museums, libraries, and communities digitize collections. Whose standards govern metadata? Which languages and classification systems structure searchable databases? Who owns digitized cultural materials and determines access conditions? Western institutions historically appropriated physical cultural artifacts; digitization risks reproducing these patterns when institutions in the Global North control digital infrastructure, standards, and access protocols for materials originating elsewhere (Christen 2012; Silverman 2015). Indigenous communities increasingly assert sovereignty over digital cultural heritage through protocols like the Local Contexts labels, which communicate traditional knowledge protections and cultural attributions that copyright law cannot capture.

Cultural sovereignty also concerns representation: whose faces, bodies, and experiences appear in training data, whose values shape algorithmic objectives, whose aesthetics determine interface design. Facial recognition systems demonstrate higher error rates for darker-skinned faces because training data over-represents light-skinned individuals (Buolamwini and Gebru 2018). Content moderation algorithms trained primarily on Global North contexts misinterpret cultural practices elsewhere, censoring legitimate expression while missing harmful content operating through local linguistic and cultural codes. Platform interfaces privilege literacy, individualism, and particular interaction styles, marginalizing oral traditions, collective decision-making, and alternative modes of engagement.

Indigenous Data Sovereignty: Epistemological Intervention

Indigenous data sovereignty movements challenge dominant technological paradigms in a novel way, beyond "adding diversity" to existing frameworks. They propose alternative epistemologies grounded in relationships, collective rights, and intergenerational responsibility (Carroll et al. 2020; Walter et al. 2021).

The CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, Ethics) assert that data about Indigenous peoples, lands, and knowledge belongs to communities, not individuals or external researchers (Carroll et al. 2020). This challenges liberal individualism's foundational premise that persons own data about themselves. Indigenous frameworks recognize that information about one community member necessarily involves others (kinship ties, land relationships, and cultural knowledge) that cannot be individualized without violating its meaning. Data sovereignty therefore requires collective governance mechanisms that Western legal and technical frameworks struggle to accommodate.

Te Hiku Media's Māori language sovereignty project is about communities building technological capacity that serves cultural continuity. The organization developed Kaitiakitanga License, adapting open-source principles to Māori concepts of guardianship (Te Hiku Media 2023). Unlike standard open licenses that permit any use, Kaitiakitanga requires users to engage Māori communities, respect cultural protocols, and ensure technology serves language revitalization rather than appropriation (Mathias 2022). This represents sovereignty through institutional innovation that encodes Indigenous values in technical and legal architecture.

Similarly, the Yawuru people's participatory data governance over the Aarli cultural database establishes community authority over digital representations of knowledge, requiring free, prior, and informed consent for any external access and prohibiting commercial use. The system embodies relational ontology: understanding knowledge as belonging to networks of relationship rather than individual creators (Kukutai and Taylor

2016). These examples show Indigenous data sovereignty offering theoretical and practical alternatives to extractive data regimes, in the form of worldbuilding.

Municipal and Community Digital Sovereignty

Cities and communities also assert cultural and social sovereignty through technological self-determination, embodying what has been identified as collective or community sovereignty: the governance of shared digital resources and common infrastructures through democratic ownership and participatory decision-making. Rather than individual users negotiating separately with platforms or states imposing top-down regulations, municipal and community initiatives create institutional forms where members collectively determine rules, share benefits, and govern according to local values and priorities. Barcelona's municipal data strategy combines technical infrastructure (city-owned sensors, open-source platforms) with participatory governance (citizen assemblies determining data use priorities) and cultural commitments (multilingual interfaces respecting Catalan alongside Spanish and English) (Calzada 2018). The approach treats data as common resource serving collective benefit rather than commodity generating private profit, demonstrating how sovereignty operates through institutional design aligning technical, legal, and normative commitments.

Community networks in rural and marginalized areas worldwide build connectivity infrastructure serving local needs rather than commercial imperatives. Projects like Zenzeleni in South Africa and TIC in Mexico demonstrate that communities can own and operate communication infrastructure, determining pricing, governance, and development priorities democratically (Metri 2017; Najera and Hasegan 2021). These networks often integrate cultural practices (community meetings determining network expansion, local languages in interfaces, pricing structures reflecting solidarity), showing how sovereignty manifests through design choices embedding community values in technical systems.

These municipal and community initiatives demonstrate sovereignty as practical capacity built through institutional design rather than merely claimed through rhetoric. They operate at scales between individual and state: neither isolated users nor national governments but communities exercising collective self-determination over technological systems mediating their social and cultural lives. This represents sovereignty's pluriversal character—multiple, nested, interoperable forms coexisting rather than a single authoritative center.

Digital connectivity can enhance the sovereignty of civil society by allowing it to organize independently of state-controlled channels. For example, during political uprisings or protests, activists often rely on encrypted messaging and social networks to coordinate (Tufekci 2017; Milan and Hintz 2013). However, these same tools can sow discord or be manipulated, as seen with disinformation campaigns and algorithm-driven polarization (Vaidhyanathan 2018). The outsized influence of a platform's design on social behavior (what content goes viral, what conversations become toxic) has led to concerns that we are losing a grip on our "cognitive sovereignty" or "attention sovereignty." (Napoli 2014; Williams 2018) This concern has spurred calls for algorithmic transparency. Some advocate for decentralized social media (like the federated Mastodon network) where communities can set their own moderation policies and algorithms (Gehl 2025). Initiatives such as community-run forums or cooperative platforms often implement democratic moderation, a microcosm of community sovereignty in action.

Design, Representation, and Power

Cultural sovereignty struggles reveal how seemingly neutral design choices encode power relations (Deepak and Manski 2025). When platforms require legal names, they marginalize Indigenous naming practices, trans individuals, and cultures where single names or community-based naming prevail. When interfaces assume nuclear family structures, they exclude extended kinship networks and chosen families. Each design choice structures whose cultural practices appear normal and whose require accommodation, explanation, or concealment.

Buddhist communities have engaged tech developers to ensure meditation apps or AI trained on Buddhist texts respect the integrity of teachings, rather than cherry-picking in a way that violates their tradition (Cook 2016; Hershock 2021). These instances show that communities are asserting control over data by designing the meaning and use of that data in line with their worldview.

Algorithmic content curation similarly exercises cultural power. When recommendation systems amplify English-language content, they make non-English creators invisible regardless of quality or relevance (Noble 2018). Platform policies developed in California moderate content globally impose particular cultural values as universal standards (Gillespie 2018; Roberts 2019). When AI-generated content replicates biases in training data, it perpetuates representational harms at scale, making marginalized groups invisible or stereotyped while centering dominant narratives.

Environmental Dimension

Data Centers and Infrastructure Footprints

Data centers exemplify digital technology's material intensity. A typical hyperscale facility operates 24/7 in buildings the size of football stadiums, consuming electricity equivalent to small cities and requiring millions of liters of water annually for cooling systems (Crawford 2021; Marx 2025). Google's global data center operations use approximately 15 billion liters of water per year; Microsoft's facilities in drought-prone regions compete with agriculture and residential use for scarce water resources (Nicolette, Ma, and Bass 2025). Yet these infrastructures employ minimal local labor (typically 20 people per facility) while demanding extensive resource inputs that host communities provide without corresponding economic benefit or decision-making authority (Marx 2025).

The geography of data centers reflects and reproduces global inequalities. Latin America, for instance, produces less than 20% of the material inputs required for data centers built on its territory, importing most components while exporting resources (energy, water, land) and providing favorable regulatory environments (Rikap et al. 2024). This pattern mirrors historical extractive relationships: peripheral regions supply raw materials while core regions capture value-added production and profits (Couldry and Mejias 2019). Data colonialism thus operates materially as well as informationally, appropriating both data and the ecological foundations enabling its processing.

Chile's experience illustrates how environmental sovereignty confronts corporate power. When the government developed a sustainability assessment tool for data center siting, industry lobbying led authorities to raise toxicity thresholds, exempting data centers from environmental reporting requirements applied to other industries (Marx 2025). Corporate sovereignty over siting and regulatory standards thus overrode territorial sovereignty over environmental protection. Communities near data centers face groundwater depletion, increased electricity costs from grid strain, and heat island effects from facility operations, yet lack mechanisms to refuse or negotiate these impacts (Crawford 2021).

Communities and nations increasingly assert environmental sovereignty over data center infrastructure. Ireland paused approvals in 2022 citing grid strain and climate concerns (Beesley 2025); U.S. towns have rejected water-intensive facilities in drought-prone regions (Nicolette, Ma, and Bass 2025); Nordic countries market themselves as 'green data havens' leveraging renewable energy and cold climates, attempting to shape global infrastructure distribution according to environmental advantages (Velkova 2016).

The narrative positioning data centers as economic development opportunities obscures their extractive character. Promised jobs materialize minimally; tax revenues remain modest; infrastructure demands often exceed contributions. One can liken them to a U.S. military base — infrastructure serving external strategic interests while imposing costs locally and constraining host sovereignty (Marx 2025). This framing captures how digital infrastructure, like military installations, operates as projection of external power.

Energy Consumption and Carbon Emissions

Digital systems' energy consumption grows exponentially, with artificial intelligence accelerating demand. Data centers account for approximately 1-2% of global electricity use (415 tWh), with projections suggesting this could reach 4% (945 tWh – equivalent to Japan's total current energy consumption) by 2030 as AI training and inference scale (International Energy Agency 2024). The sovereign AI race is a critical factor here: nations building domestic AI capabilities require massive computational infrastructure. Training state-of-the-art language models that serve potentially billions of users multiplies environmental costs (Chavez 2024; Schiffer and Matzakis 2025). Cryptocurrency mining compounds the problem, with Bitcoin alone consuming more electricity than many nations (Greenberg and Bugden 2019; Krause 2024). These exponentially growing energy demands raise sovereignty questions about resource allocation and climate commitments.

These aggregate figures implicitly ask political questions: Who controls energy-intensive computational resources? Whose priorities justify massive energy consumption? Can communities refuse energy-consuming infrastructure when it conflicts with climate commitments or renewable energy goals? Environmental sovereignty requires authority to make these determinations collectively.

Renewable energy powering data centers, while better than fossil fuels, does not resolve sovereignty issues. When tech companies secure renewable energy contracts, they often purchase output from projects that would otherwise serve local grids, effectively diverting clean energy to corporate use while fossil fuel generation continues elsewhere (Pellow 2007). Community energy sovereignty (collective ownership and governance over energy systems serving local priorities) confronts corporate purchasing power and utility industry structures favoring large-scale contracts over distributed, locally controlled alternatives.

Electronic Waste and Circular Economy

Device manufacturing and disposal create severe environmental and health impacts. Electronic waste contains toxic materials (lead, mercury, cadmium) that contaminate soil and water when improperly disposed. Approximately 62 million metric tons of e-waste were generated globally in 2022, with only 22.3% formally recycled. Projections suggest this could reach 82 million tons by 2030 (Baldé 2024). The Global E-waste Monitor shows waste generation growing five times faster than formal recycling capacity, creating mounting environmental and health crises.

Much e-waste flows from wealthy nations to Global South communities where informal recycling exposes workers to toxic substances without protective equipment or health support (Pellow 2007; Lepawsky 2018). This “environmental racism” embeds sovereignty inequalities: Global North consumers enjoy device convenience while Global South communities bear disposal burdens.

Some African countries (like Ghana) are simultaneously developing their own e-waste recycling industries, aiming to turn what's dumped on them into resources (metals recovery) in a controlled way (Grant and Oteng-Ababio 2019; Ossie 2024). This should be read as a pragmatic sovereignty strategy to mitigate damage and reclaim value.

Planned obsolescence accelerates waste generation. Manufacturers design devices with limited lifespans through non-replaceable batteries, proprietary components preventing repair, and software updates rendering older models unusable. This serves profit maximization but contradicts environmental sustainability and community sovereignty. When devices cannot be repaired locally, communities depend on manufacturers for maintenance or replacement, surrendering control over technological lifecycles.

Right-to-repair movements assert sovereignty through demanding legislative changes compelling manufacturers to provide repair documentation, tools, and parts based on a conception of personal property (Perzanowski and Schultz 2018). The EU's eco-design directives and right-to-repair regulations represent regulatory sovereignty challenging corporate control over device longevity (European Commission 2024). Yet enforcement confronts lobbying, technical complexity, and manufacturers' global operations enabling regulatory arbitrage. Environmental sovereignty thus requires not just laws but technical capacity to implement repairs and institutional power to compel manufacturer cooperation.

Circular economy approaches propose systemic redesigns prioritizing durability, repairability, and material recovery. Fairphone, a Dutch social enterprise, demonstrates that devices can be designed with modular, replaceable components enabling user repairs and extended lifespans. The company's smartphones feature easily replaceable batteries, screens, and cameras, with spare parts sold directly to consumers and detailed repair guides published openly. They challenge planned obsolescence while sourcing conflict-free minerals and ensuring fair labor conditions, even in the face of discouraging market logic (Proske et al. 2020). However, such initiatives struggle against dominant business models favoring disposability and network effects concentrating market power in proprietary ecosystems. Transitioning to circular models requires coordinated regulatory action, public procurement favoring sustainable design, and technical infrastructure supporting repair and refurbishment. These are all sovereignty exercises demanding collective capacity.

Mineral Extraction and Supply Chains

Rare earth minerals essential for device manufacturing come from ecologically and socially destructive operations: cobalt mining in Congo's Katanga province employs 40,000 children in artisanal mines facing toxic exposure; lithium extraction in Chile's Atacama consumes 500,000 liters of water per ton, devastating Indigenous Atacameño communities' aquifers (Amnesty International 2023; Kalantzakos 2020). Rare earth refining concentrates in China's Inner Mongolia, where mining operations have created toxic lakes contaminating land and water for nearby Mongolian herding communities (Klinger 2017).

At each stage (extraction, refining, manufacturing, consumption, disposal) communities bearing environmental and health costs capture minimal economic value while peripheral regions supply resources enabling core technological advancement. Mining communities in Congo lack electricity despite producing cobalt powering global digital devices. Atacameño peoples lose water resources so distant consumers can charge smartphones. This represents digital sovereignty disconnected from supply chain justice: neo-colonial relationships where peripheries provide resources while cores enjoy benefits and externalize costs. Environmental sovereignty requires transforming these relationships, ensuring resource-providing communities exercise authority over extraction decisions and capture equitable value from minerals extracted from their lands.

Indigenous communities asserting sovereignty over lands containing minerals confront pressures to permit extraction for "global technological needs." Yet framing extraction as necessary for digital progress doesn’t tell us whose technologies, serving whose interests, justified by whose priorities. Environmental sovereignty means the authority to refuse extraction, to determine whether mineral resources support community wellbeing or corporate accumulation, and to negotiate terms ensuring benefits exceed costs. Current arrangements typically subordinate Indigenous sovereignty to state interests promoting mining investment, demonstrating how territorial sovereignty at national scale can undermine community sovereignty at local scale (Thi and Esplen 2023).

Embodied and Cumulative Impacts

Beyond headline issues like energy consumption and e-waste, digital technologies generate diffuse environmental and health impacts. Electromagnetic frequencies from wireless infrastructure raise health concerns for sensitive individuals, while LED screen proliferation disrupts circadian rhythms, affecting sleep and wellbeing. Manufacturing processes release particulates and chemicals into air and water, while data center heat emissions create urban heat islands (The Investor's Podcast Network 2025).

These embodied impacts are difficult to measure, regulate, or attribute to specific sources, yet cumulatively shape environmental health. Environmental sovereignty at this level requires precautionary approaches: authority to refuse technologies when impacts remain uncertain, capacity to monitor environmental health indicators, and mechanisms to hold technology producers accountable for cumulative harms.

Sustainable digital design is a vital aspect. Communities are designing energy-efficient software requiring less server time, modular repairable hardware extending device lifespans, and 'carbon-aware' services scheduling processing when renewable energy is abundant (Hilty and Aebischer 2015; Widdicks et al. 2022). These innovations tie back to control of technical parameters, enabling environmental sovereignty.

Local communities building digital infrastructure can inherently incorporate sustainability. The 'Internet Roshni' project in rural India is an excellent illustration: tea garden communities established local Wi-Fi networks for tribal members, approaching sustainability holistically across financial, technical, social, cultural, and environmental dimensions (Kumari and Mukhopadhyay 2024). They selected low-power equipment with minimal radiation, powered by solar panels. By minimizing energy use and e-waste while leveraging renewable energy, the community exercised environmental sovereignty. Their network expansion served connectivity needs without harming local environments or creating unsustainable cost burdens.

The discourse of digital technology as "clean" or "green" compared to industrial manufacturing conceals these material realities. While digital systems may generate less visible pollution than factory smokestacks, their environmental footprints are substantial, globally distributed, and growing rapidly. Environmental sovereignty demands transparency about impacts across entire technological lifecycles, beyond narrow metrics such as operational emissions.

Ecological Internationalism and Shared Governance

Environmental challenges transcend territorial boundaries, requiring coordination that traditional sovereignty frameworks struggle to accommodate. Climate change, e-waste flows, and supply chain impacts operate globally; no nation acting alone can secure environmental protection. This motivates ecological internationalism: coordinated action addressing shared challenges while respecting diverse approaches and ensuring equitable burden-sharing (Rikap et al. 2024).

The Democratic and Ecological Digital Sovereignty Coalition proposes international cooperation on public digital infrastructure, shared research agendas, and coordinated environmental standards (Rikap et al. 2024. As an alternative to nationalist technology competition driving redundant infrastructure and resource waste, ecological internationalism envisions pooled resources, interoperable systems, and distributed governance. This represents sovereignty as networked cooperation, recognizing that environmental protection requires collective action.

Global South countries assert they should not bear digitalization's environmental costs without enjoying benefits. Movements pressure for extended producer responsibility, compelling tech companies to finance electronic take-back and recycling regardless of disposal location (Atasu and Van Wassenhove 2012). There are also calls for tech companies to invest in environmental restoration, asserting that the global digital economy must respect local sovereignties. Within countries Indigenous communities demand consultation authority over infrastructure projects: Pacific Northwest tribes have contested data centers on ancestral lands, asserting sovereignty means deciding whether projects align with environmental stewardship commitments (Whyte 2018).

One could foresee 'climate data sovereignty' becoming a theme: ensuring that communities vulnerable to climate change have control over data about their environment (e.g. using Indigenous knowledge in climate models with respect and consent) (Crate and Nuttall 2016; Whyte, Brewer, and Johnson 2016). This would merge cultural, environmental, and technical sovereignty in a powerful way.

Yet power asymmetries complicate cooperation. When wealthy nations propose environmental standards, they may impose costs on poorer nations developing digital capacity. International agreements may favor incumbents over newcomers building alternatives. Environmental sovereignty therefore requires equitable governance ensuring Global South nations, Indigenous peoples, and marginalized communities participate as decision-making equals.

Ultimately, the environmental dimension shows that digital sovereignty without ecological consciousness serves continued extraction, and ecological sustainability without sovereignty leaves communities powerless. The two must be pursued together.